

# Floor van Donkelaar

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Office K23  
Institute of Astronomy  
University of Cambridge

*Main research interests:* The formation and evolution of galactic components in disc galaxies at high redshift and the role of stellar clusters by means of (cosmological) simulations.

*Languages:* Dutch (Native), English (Fluent), German (Elementary), Swedish (Elementary)

## APPOINTMENTS

**Herchel Smith Postdoctoral Research Fellow** Oct 2025 - Present  
University of Cambridge, United Kingdom  
*Institute of Astronomy (IoA) & Kavli Institute for Cosmology (KICC)*

## EDUCATION

**PhD in Computational Astrophysics** Sep 2021 - Sep 2025  
University of Zurich, Switzerland  
*Thesis:* [Disc Galaxies and Their Components at High Redshift: Similarities with the Local Universe](#)  
*Supervisor:* Prof. Dr. Lucio Mayer

**MSc in Astrophysics** Aug 2019 - May 2021  
Lund University, Sweden  
GPA: 3.83/4.00  
*Thesis:* [The Fate of Stars Born in Gas-Rich High Redshift Galaxies](#)  
*Supervisor:* Prof. Dr. Oscar Agertz

**BSc in Sociotechnical Engineering, Minor in Modern Physics** Sep 2016 - Jul 2019  
University of Twente (ATLAS), The Netherlands  
GPA: 3.85/4.00  
*Thesis:* The Star Formation Rate, Metallicity and Thermal Pressure in Galaxies at  $z=0.4$  using MUSE  
*Supervisor:* Dr. Kasper Borello Schmidt from *Leibniz-Institut für Astrophysik Potsdam*

## GRANTS & FELLOWSHIPS

**Herchel Smith Postdoctoral Research Fellowship** ( $\sim$  £185K) 2025

## EARLY RESEARCH EXPERIENCE

**Oxford University**  
Research Internship Jun 2020 – Nov 2020  
*Supervisor:* Dr. Kearn Grisdale  
Analyzing GMCs in the Large Magellanic Cloud with the use of N-body hydrodynamical simulations.

**Leibniz-Institut für Astrophysik, Potsdam**  
Summer Project Jul 2017  
*Supervisor:* Dr. Kasper Borello Schmidt  
Generating template spectra of MUSE-Wide emission lines sources.

## TEACHING

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### IoA Summer Internship Programme Postdoc Seminar

From the First Star Clusters to Today's Globular Clusters

Aug 2026

### Astrophysics Part II Supervisor, University of Cambridge

Topics in Astrophysics

Lent Term 2026

### Teaching Assistant, University of Zurich

AST 202: The Universe: Contents, Origin, Evolution and Future

Spring 2025, 2024 & 2022

AST 201: Introduction to Astrobiology

Fall 2024 & 2021

AST 245: Computational Astrophysics

Fall 2024 & 2023

AST 295: Astrobiology proseminar

Fall 2023 & 2022

AST 248: The Sun and Planets

Spring 2023

## SUPERVISING & MENTORING

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### Master's project

#### Co-supervisor:

Daniel Swinger (ETH Zurich)

2025 - 2026

Xiaojie Liao (University of Lancashire)

2026

### Bachelor's project

#### Supervisor:

Romy Jager (University of Twente)

2026

## (PUBLIC) OUTREACH

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### Co-chair **IWD 2026**, University of Cambridge

Dec 2025 - Present

International Women's Day (IWD) organises a series of events within the Institute of Astronomy to celebrate the achievements of women astronomers and to recognize the challenges that they still face within the field.

### Reviewer for MNRAS, A&A

Oct 2024 - Present

### Chief Public Relations, Green Team Twente

Jun 2017 - Sep 2018

Led PR and design for a student team developing one of the world's most efficient hydrogen city cars. Managed branding, media, and events, contributing to winning the 2018 Shell Eco-Marathon Communication Award.

### Member Faculty Council EEMCS, University of Twente

Sep 2017 - Aug 2018

Advised the faculty management team on policies affecting staff and students, advocating for their interests.

### Workshop developer, University of Twente

May 2017 - Jun 2018

Supported and developed workshops about mentoring and the writing of personal development plans in the science track of the honours program at the University of Twente.

## PUBLICATION OVERVIEW

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Number of (first-author) publications:

8 (7)

Number of (first-author) submissions:

2 (2)

### List of Publications

1. "*Cosmic Wallflowers: The circumgalactic origins of isolated ultra-compact star clusters at  $z > 7$* ", **van Donkelaar F.**, Mayer L., Capelo P. R., et al., 2026 *MNRAS*, 548, x

2. *"Introducing the PHOEBOS simulation: galaxy properties at the dawn of galaxy formation"*, van Donkelaar F., Capelo P. R., Mayer L., et al., 2025 [MNRAS](#), **543**, 2760
3. *"In-situ formation of primordial star clusters at  $z > 7$  via gaseous disc fragmentation; shedding light on the gems and on rapid black hole growth in the early Universe"*, Mayer L., van Donkelaar F., Messa, M., et al., 2025, [ApJL](#), **981**, 8
4. *Exploring the fate of primordial discs in Milky Way-sized galaxies with the GigaEris simulation"*, van Donkelaar F., Mayer L., Capelo P. R., et al., 2025, [MNRAS](#), **539**, 1259
5. *"Wandering intermediate-mass black holes in Milky Way-sized galaxies in cosmological simulations: myth or reality?"*, van Donkelaar F., Mayer L., Capelo P. R., et al., 2025, [MNRAS](#), **538**, 2255
6. *"Stellar cluster formation in a Milky Way-sized galaxy at  $z > 4$  - II. A hybrid formation scenario for the nuclear star cluster and its connection to the nuclear stellar ring"*, van Donkelaar F., Mayer L., Capelo P. R., et al., 2024, [MNRAS](#), **529**, 4104
7. *"Stellar cluster formation in a Milky Way-sized galaxy at  $z > 4$  - I. The proto-globular cluster population and the imposter amongst us"*, van Donkelaar F., Mayer, L., Capelo, P. R., et al. 2023, [MNRAS](#), **522**, 1726
8. *"From giant clumps to clouds - II. The emergence of thick disc kinematics from the conditions of star formation in high redshift gas rich galaxies"*, van Donkelaar F., Agertz, O., & Renaud, F. 2022, [MNRAS](#), **512**, 3806

## SELECTED TALKS

A full list of the talks I have given can be found on my [website](#).

- **[Invited]** *'Clustered Star Formation at High Redshift: A Simulation Perspective'*  
EAS 2026: Stellar clumps and star clusters as building-blocks of galaxy assembly (SS5), 4 July 2026
- *'Disc Fragmentation and Cluster Formation in Clumpy High-Redshift Galaxies'*  
BeyondHubble: Revisiting the Hubble Sequence Across Cosmic Time (Cambridge), 22 April 2026
- *'Simulating the Early Universe: Shedding Light on Clusters seen by JWST at  $z > 4$ '*  
Sexton Center for Astrophysics Riccardo Giacconi, 1 Jul 2025
- *'First Results of the Phoebos Simulation: Galaxy Sizes during the Early Universe'*  
SKAO Council meeting (Zurich), 17 Mar 2025
- *'The GigaEris Simulation: What makes a high redshift (disc) galaxy different?'*  
Northwestern University, 26 Sep 2024
- *'The GigaEris Simulation: Stellar clusters in MW-sized galaxies at  $z > 4$ '*  
Flatiron Institute, CCA (New York), 13 Sep 2024
- **[Invited]** *'Stellar Systems at  $z > 4$ : The hybrid formation scenario for the nuclear star cluster'*  
Lund University, 8 May 2023
- *'The Formation of the Nuclear Star Cluster'*  
University of Chicago, 28 Mar 2023
- *'On disc kinematics: The influence of the thin disc on star clusters'*  
Aspen Center for Physics, 18 Mar 2022

SCIENTIFIC MEETINGS & WORKSHOPS

**Zoom-in Views of galaxy Disks Across Cosmic Time: Bridging Simulations and Observations**  
Nordita, Stockholm, 1 - 19 Jun 2026

**NBIA Workshop on Open Problems in Astrophysical Dynamics**  
Copenhagen, 10 - 13 Jun 2025

**“SKA1-MID Observing Capability at High Frequencies” Meeting**  
ISSI, Bern, 22 Jun 2023

**Star Formation Within Evolving Galaxies: The Revolution of Upcoming Space Missions**  
ISSI, Bern, 29 Aug – 2 Sep 2022 & 28 Aug – 1 Sep 2023

SCIENTIFIC MEMBER

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|---------------------------------------------------------------------------------|----------------|
| <a href="#">Research Associate</a> at Emmanuel College, University of Cambridge | 2026 - Present |
| Member of the HI Galaxy Science working group of SKA                            | 2022 - Present |

AWARDS

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|------------------------------------------------------------------------|------|
| University College Twente Third-year Award, University College Twente  | 2019 |
| More Than A Degree Awards, University of Twente                        | 2018 |
| Communication Award Shell Eco-Marathon, Shell                          | 2018 |
| University College Twente Second-year Award, University College Twente | 2018 |